Threshold Attention Network for Semantic Segmentation of Remote Sensing Images

Wei Long, Yongjun Zhang, *Member, IEEE*, Zhongwei Cui, Yujie Xu, and Xuexue Zhang

***Abstract*— Semantic segmentation of remote sensing images is essential for various applications, including vegetation monitoring, disaster management, and urban planning. Previous studies have demonstrated that the self-attention mechanism (SA) is an effective approach for designing segmentation networks that can capture long-range pixel dependencies. SA enables the network to model the global dependencies between the input features, resulting in improved segmentation outcomes. However, the high density of attentional feature maps used in this mechanism causes exponential increases in computational complexity. In addition, it introduces redundant information that negatively impacts the feature representation. Inspired by traditional threshold segmentation algorithms, we propose a novel threshold attention mechanism (TAM). This mechanism significantly reduces computational effort while also better modeling the correlation between different regions of the feature map. Based on TAM, we present a threshold attention network (TANet) for semantic segmentation. The TANet consists of an attentional feature enhancement module (AFEM) for global feature enhancement of shallow features and a threshold attention pyramid pooling (TAPP) module for acquiring feature information at different scales for deep features. We have conducted extensive experiments on the international society for photogrammetry and remote sensing (ISPRS) Vaihingen and Potsdam datasets. The results demonstrate the validity and superiority of our proposed TANet compared with most state-of-the-art models.**

***Index Terms*— Remote sensing images, self-attention mechanism (SA), semantic segmentation, threshold attention mechanism (TAM), threshold attention network (TANet).**

I. Introduction

REMOTE sensing is an important source of geospatial information and plays a crucial role in numerous applications, including urban planning [1], [2], [3], vegetation monitoring [4], [5], military surveillance [6], disaster monitoring [7], and meteorological monitoring [8]. One of the fundamental tasks in remote sensing is semantic segmentation, which involves assigning a unique category label to each pixel in an image.

Deep learning is now widely employed in various red green blue (RGB) image processing tasks. FCN [9] was the first fully convolutional network proposed and used in the field of semantic segmentation, implementing end-to-end pixel-level semantic segmentation. Since then, numerous networks with improvements over FCN have been proposed, including U-Net, pyramid scene parsing network (PSPNet), the DeepLab series networks, and statistical texture learning network (STLNet). These networks typically have a two-part structure, consisting of an encoding and a decoding component. Despite the improved effectiveness of semantic segmentation achieved by these encoding–decoding network models, two important challenges still remain.

First, the downsampling operation within the encoding component of a network model often leads to the loss of fine information in the original image, resulting in coarse and inaccurate predictions [10]. Specifically, in regions with rich detail, such as object boundaries, the predictions tend to be particularly poor. To address this issue, a common strategy is to integrate low-level features with rich edge information into high-level features that contain more semantic information [11], [12], [13], [14], [15]. This enhances the accuracy of the final prediction results.

Furthermore, convolutional operators in convolutional neural networks possess limited capability to capture long-range dependencies due to their emphasis on local features and close relationships [16]. The size of the receptive field provides an estimation of the amount of contextual information that can be obtained. However, the receptive field of fully convolutional networks only increases linearly with the depth of the network [10].

To capture more distant dependencies in the feature map, Yu and Koltun [17] introduced the concept of dilated convolution, enabling the exponential growth of the receptive field without sacrificing resolution. Chen et al. [18] further proposed an atrous spatial pyramid pooling (ASPP) module based on multiscale dilated convolution to extract the feature information of objects at different scales. However, the use of dilated convolutions and stacked convolutional layers only provides limited contextual information, leading to limitations in modeling dependencies between distant pixels in the feature map [19].

The self-attention mechanism (SA) has been extensively employed in tasks such as natural language processing and computer vision, owing to its potent ability to capture long-range dependencies. A prominent example of this is the non-local network proposed by Wang et al. [20], which calculates attention weights between pixels at different locations through dot-product operations on feature maps. This integration of self-attention into the convolutional neural network enables the

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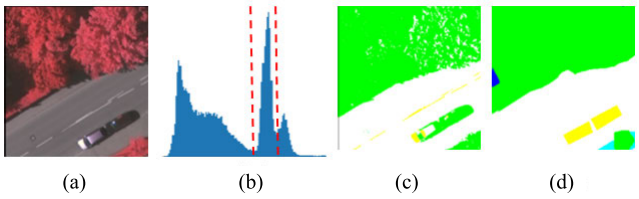


Fig. 1. Traditional threshold segmentation method. (a) Is the original remote sensing image, (b) is the image histogram of (a) in the red channel, (c) is the segmentation result map obtained by the traditional threshold segmentation method, and (d) is the label map of (a).

model to effectively capture the relationships between distant pixels.

However, this SA has two obvious limitations. First, generating a dense attentional feature map requires quantifying the correlation between every pixel pair, resulting in computational complexity. Second, neighboring pixels in remote sensing images are often highly correlated, and their dependencies play a significant role in extracting semantic information. However, the SA equally considers all dependencies between pixel pairs when calculating relationships. This not only disregards local information but also introduces redundant attention weights [21], resulting in a detrimental effect on feature representation [19].

As illustrated in Fig. 1, the segmentation map Fig. 1(c) can be efficiently obtained by setting two threshold values (represented by the red dotted line) for the histogram of the red channel of the image presented in Fig. 1(b). The traditional threshold segmentation method possesses the advantage of aggregating all pixels with similar values across the entire image, yielding detailed edge information. However, it lacks semantic information, which leads to object misclassification, such as cars in Fig. 1(c) being misidentified as trees. In contrast, both neural networks and SAs demonstrate strong semantic information extraction capabilities.

In the process of segmenting objects in images, humans often divide the images into numerous pixel-based regions. Consequently, it is only necessary to consider the inter-block pixel relationships, rather than the relationships between individual pixels. Based on this idea, we propose the TAM. Input features are initially subjected to global quantization based on specific threshold values, generating a global threshold information vector. This vector undergoes a series of convolutional and dot-product operations to compute the attention weight matrix for similar pixel aggregation regions. The final threshold attention weights are obtained by restoring the location information through another path. Compared with the SA, TAM not only significantly reduces computational complexity but also effectively addresses the issue of redundant dependencies between pixel pairs negatively impacting feature representation.

The primary contributions of this study are as follows.

1) We introduce a novel TAM that focuses on the dependencies of pixel regions rather than pixel pairs. This attention mechanism provides a linear kernel attention computational complexity and effectively models the correlation between similar regions in the feature map.
2) We develop an attentional feature enhancement module (AFEM) based on TAM. The AFEM modules can significantly improve the feature information of various regions where each category is located, thus obtaining an output with richer detailed features and clearer segmentation boundaries, which is beneficial for refining deeper features.
3) We have improved the ASPP module by integrating the TAM module and the enhanced ASPP module, resulting in the TAPP module. This integration enables the model to effectively capture rich global contextual information, multiscale information, and model the relationships between similar pixel regions.
4) We propose a novel threshold attention network (TANet). The TANet consists of two key components: the AFEM module and the TAPP module. The AFEM module is responsible for enhancing the shallow features obtained from the image. These shallow features are then fused with deep features enhanced by the TAPP module. The resulting segmentation map is both semantically rich and finely detailed.

II. Related Work

A. Semantic Segmentation

Semantic segmentation, which assigns semantic labels to each pixel in an image, plays a vital role in image processing. Traditional approaches to semantic segmentation often yield suboptimal accuracy. Nevertheless, the advent of deep neural networks has facilitated considerable advancements in semantic segmentation accuracy due to their capacity for automatic extraction of more informative image features and integration of richer contextual information. Consequently, most state-of-the-art semantic segmentation algorithms presently utilize deep neural networks as their foundation.

The FCN [9] was a pioneering CNN architecture that effectively performed end-to-end semantic segmentation. Subsequent to its introduction, numerous methods have been developed that build upon the innovations of FCN. For instance, U-Net [11] introduced a symmetric encoder–decoder structure, where the encoding component extracts image features, and the decoding component recovers the edge details lost during downsampling. The ASPP module, incorporated in DeepLab [18], enhances the ability of the network model to capture contextual information. STLNet [22] leverages statistical analysis of global low-level information in feature maps to effectively extract statistical texture features at multiple scales, thus enhancing texture details. Guo et al. [23] reevaluated the characteristics necessary for successful semantic segmentation. They proposed SegNeXt, a novel convolutional attention network that utilizes inexpensive convolutional operations and achieves performance superior to transformer-based models.

Xu and Jiang [24] proposed the high-resolution boundary constraint and context augmentation network (HBCNet), which utilizes boundary information, semantic information of categories, and regional feature representations to improve semantic segmentation accuracy. CTMFNet [25], a multiscale fusion network, employs an encoder–decoder framework that integrates CNN and transformer mechanisms into its backbone

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network. To effectively combine local and global information in the dual backbone encoder, the authors propose a dual backbone attention fusion module (DAFM). The decoder comprises a multilayer densely connected network (MDCN), which bridges the semantic gap between scales.

B. Self-Attention Mechanism

SA, initially employed in the domain of NLP, has since been adopted in various other fields. Vaswani et al. [26] combined an SA with a Recurrent Neural Network, allowing the network to focus on key image locations. Another notable contribution is the work of Wang et al. [20], who proposed a nonlocal approach using an SA to model interdependence between input feature map pixels.

There are two significant limitations associated with the SA. First, as the resolution of the input image increases, the computational burden on the network also becomes significantly large. Second, the SA simply computes a dot product on the matrix that encompasses all feature information, which does not constitute a robust representation of the features.

Several studies have aimed to enhance the efficiency and effectiveness of SAs. One example is the criss-cross network (CCNet) [27], which employs a crossover attention mechanism to compute long-range dependencies with reduced computational cost. Another study [28] introduced a relation-aware global attention (RGA) module to better learn attention weights by incorporating global structural information. Sun et al. [29] proposed a successive pooling attention network (SPANet) with a successive pooling attention module (SPAM) that pools the value matrix to obtain features at different scales, leading to multiscale attentional feature extraction. Guo et al. [30] proposed a novel attention mechanism referred to as "External Attention." It incorporates two external, trainable memory modules that compute long-range dependencies between sample features to obtain attention.

We propose a novel threshold attention mechanism (TAM), which differs from prior methods in its focus on dependencies among pixel regions rather than pixel pairs. TAM aggregates features within different thresholds and applies attention to regions after aggregation. This integration of traditional threshold segmentation into self-attention reduces computational complexity and eliminates redundant noise information in the attention matrix.

C. Scaling Attention Mechanism

In addition to SAs, other scaled attention mechanisms have the capacity to automatically learn attention weights during the training phase, assessing the relevance of channel or spatial features. For instance, the squeeze-and-excitation (SE) module in SE network (SENet) [31] is employed to adaptively model the interdependencies between the channels of the feature map, and then the original input feature map is recalibrated based on the weights obtained for each channel. Convolutional block attention module (CBAM) [32] and bottleneck attention module (BAM) [33] both model attentional weights for a given intermediate feature map in a network along both spatial and channel dimensions. However, they differ in the way they combine these weights; CBAM combines them in series, whereas BAM combines them in parallel. Li et al. [34] designed a new kernel attention mechanism with linear complexity to alleviate the large computational requirements in the attention mechanism. They proposed a multiattention network (MANet), which can combine the local feature maps extracted by the backbone network with global dependencies to adaptively weight the interdependent channel maps.

Our proposed AFEM encompasses a Channel Attention Module to dynamically learn the correlation between the feature map channels and weight coefficients. This weighting approach allows AFEM to automatically differentiate the importance of the different channels of the input features. It assigns greater weight to the more significant channels, which are crucial for achieving enhanced, detailed features.

III. Methodology

A. Overview

The TANet is introduced with its overall structure in Fig. 2. For feature extraction during the encoding phase, the backbone network is ResNet101 with dilated convolution. The shallow network output provides abundant details but lacks semantic information, whereas the deep network output offers rich semantic information but lacks details. The three shallow feature maps obtained from the encoding phase are concatenated and fed into the AFEM to get a feature map with enhanced detailed information. The deep features from the backbone network are input into the TAPP to obtain a feature map with rich semantic and contextual information. These complementary feature maps are then concatenated and fused to get a unified feature map. Finally, bilinear interpolation is utilized to obtain the ultimate prediction output map, which possesses the same dimensions as the input image.

The TANet integrates global contextual information and feature information at various scales to produce high-level features enriched with semantic information. Moreover, TANet also improves low-level features, which are replete with detail but devoid of semantic information. The progressive fusion of these two types of features results in a more precise and detailed segmentation prediction map.

B. Threshold Attention Mechanism

The threshold segmentation method is a widely used algorithm in conventional image segmentation. This method is based on the principle that the pixel values of different objects in an image are significantly distinct. To obtain the required pixel thresholds, a calculation can be performed or the pixel statistical histogram of the image can be processed. Subsequently, the pixels in the image are classified based on these thresholds, resulting in the segmentation of the different objects present in the image.

Inspired by this traditional approach, we present the TAM, which learns attention weights for different pixel regions in a feature map. To achieve this, we quantize each channel of the input feature map with a global threshold information matrix, resulting in a threshold feature matrix. This matrix undergoes convolutional and dot-product calculations to obtain an attention weight matrix. The input feature map is also discretized into feature classes to form a position matrix. By multiplying

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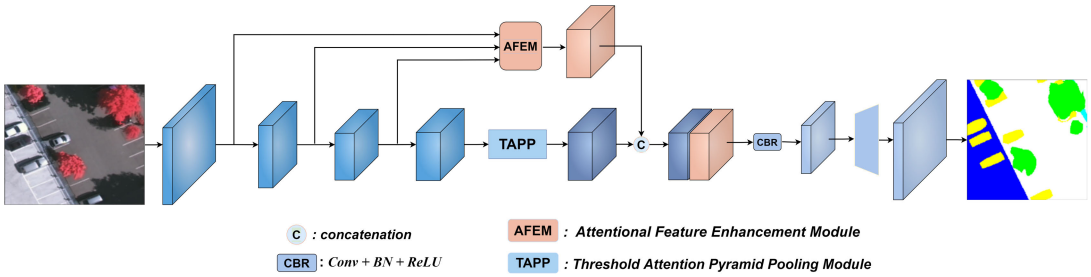


Fig. 2. TANet utilizes the ResNet101 backbone network to extract features. In addition, it employs the AFEM module to enhance the feature information obtained from the shallow network, and the TAPP module to capture rich global semantic information from the deep features. Subsequently, the two complementary feature maps are integrated to acquire a consolidated feature map. Ultimately, bilinear interpolation is leveraged to generate the ultimate predicted output map.

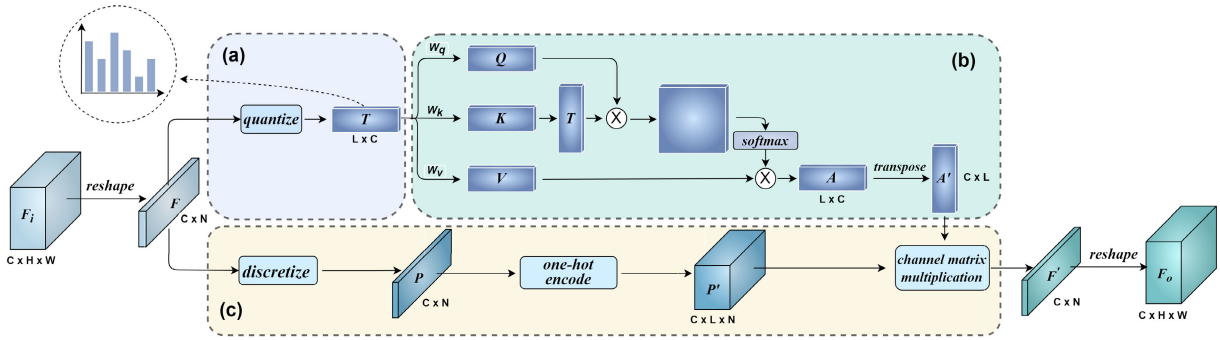


Fig. 3. TAM consists of three parts: (a) for thresholding the input features, (b) for calculating the attention weight matrix, and (c) for recovering location information for features. TAM is an attention mechanism that exhibits linear computational complexity and effectively models the correlation between similar regions in the feature map.

the attention weight matrix and position matrix, we get an output matrix that assigns attention weights to different pixel regions. Fig. 3 depicts a graphical representation of TAM.

1) Thresholding the Input Features: Define the input features as $\boldsymbol{F}_i \in \mathbb{R}^{C\times H\times W}$ and then reshape the features into $\boldsymbol{F} \in \mathbb{R}^{C\times N}$, where $N = H \times W$. Quantize each channel in $\boldsymbol{F}$ separately using a threshold that is based on the feature data distribution in the different channels. This quantization enables the grouping of pixels with similar characteristics in the original feature into disparate threshold clusters

$$\boldsymbol{T}_{c,l} = \frac{\max(\boldsymbol{F}_c) - \min(\boldsymbol{F}_c)}{2\boldsymbol{L}} \times (2l-1) + \min(\boldsymbol{F}_c) \quad (1)$$

where $c \in [1, C]$ and $l \in [1, L]$. $\boldsymbol{F}_c$ denotes the feature data of the cth channel in matrix $\boldsymbol{F}$. $\boldsymbol{L}$ represents the number of feature levels to be quantized, which implies that the data of every channel will be divided into $\boldsymbol{L}$ intervals of equal size based on a certain threshold. $\boldsymbol{T}_{c,l}$ denotes the quantization result when the feature of the cth channel in the input matrix F is within the lth threshold, and the feature matrix $\boldsymbol{T} \in \mathbb{R}^{L\times C}$ is obtained after this quantization operation.

2) Calculating the Attention Weight Matrix: Similar to the dot-product attention, we use three different projection matrices, $\boldsymbol{W}_q \in \mathbb{R}^{C\times C}$, $\boldsymbol{W}_k \in \mathbb{R}^{C\times C}$, and $\boldsymbol{W}_v \in \mathbb{R}^{C\times C}$, to generate the corresponding query matrix $\boldsymbol{Q}$, key matrix $\boldsymbol{K}$, and value matrix $\boldsymbol{V}$

$$\boldsymbol{Q} = \boldsymbol{T}\boldsymbol{W}_q \in \mathbb{R}^{L\times C} \quad (2)$$

$$\boldsymbol{K} = \boldsymbol{T}\boldsymbol{W}_k \in \mathbb{R}^{L\times C} \quad (3)$$

$$\boldsymbol{V} = \boldsymbol{T}\boldsymbol{W}_v \in \mathbb{R}^{L\times C} \quad (4)$$

$$\rho\left(\boldsymbol{Q}\boldsymbol{K}^T\right) = \text{softmax}_{\text{row}}\left(\boldsymbol{Q}\boldsymbol{K}^T\right). \quad (5)$$

We employ a normalization function ρ to measure the similarity between the ith query feature $\boldsymbol{q}_i^T \in \mathbb{R}^C$ and the jth key feature $\boldsymbol{k}_j \in \mathbb{R}^C$, i.e., $\rho(\boldsymbol{q}_i^T \cdot \boldsymbol{k}_j) \in \mathbb{R}^1$. This matrix $\boldsymbol{Q}\boldsymbol{K}^T$ models the dependencies between different threshold features (features in different pixel regions) for different channels in the threshold feature matrix $\boldsymbol{T}$. Obtain the attention matrix by first normalizing the attention weight values in the relationship matrix $\boldsymbol{Q}\boldsymbol{K}^T$ via the softmax function (denoted by $\text{softmax}_{\text{row}}$). Then, generate the attention matrix by reweighting V with the normalized attention weight values

$$\boldsymbol{A} = \rho\left(\boldsymbol{Q}\boldsymbol{K}^T\right)\boldsymbol{V}. \quad (6)$$

3) Recovering Location Information for Features: Matrix $\boldsymbol{A}$ holds global dependency information from input matrix $\boldsymbol{F}$ but lacks corresponding location information for each pixel feature due to prior quantization. As shown in Fig. 3, in order to retrieve this information, a "discretize" operation is performed on matrix F to obtain matrix $\boldsymbol{P}$, which records the quantization levels for all pixel features. This allows for the reconstruction of the original pixel location information in input feature $\boldsymbol{F}$

$$\boldsymbol{P}_{c,n} = \left\lfloor \frac{\boldsymbol{F}_{c,n}}{\boldsymbol{L}} \right\rfloor \quad (7)$$

where $n \in [1, N]$. $\boldsymbol{F}_{c,n}$ is the nth feature of the cth channel of the input feature matrix $\boldsymbol{F}$. $\boldsymbol{P}_{c,n}$ is the integer feature value obtained by thresholding $\boldsymbol{F}_{c,n}$ according to $\boldsymbol{L}$.

From this, we can obtain a matrix $\boldsymbol{P} \in R^{C\times N}$ that records the location information of the corresponding quantization level

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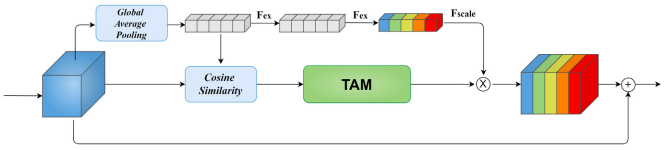


Fig. 4. Structure of the AFEM is composed of three branches: the first one acquires channel attention, the second one enhances threshold attentional features, and the third one provides residual connectivity.

of each pixel feature of the matrix $\boldsymbol{F}$. The matrix $\boldsymbol{P}$ is then one-hot encoded to obtain the matrix $\boldsymbol{P}'$, and the matrix $\boldsymbol{A}$ is transposed to obtain $\boldsymbol{A}'$

$$\boldsymbol{F}'_c = \boldsymbol{A}'_c \cdot \boldsymbol{P}'_c \tag{8}$$

where $\boldsymbol{A}'_c \in \mathbb{R}^{1\times L}$ and $\boldsymbol{P}'_c \in \mathbb{R}^{L\times N}$ are the vectors of the cth channel in the matrices $\boldsymbol{A}'_c$ and $\boldsymbol{P}'_c$, respectively. The threshold attention weights are reassigned to the features by multiplying the $\boldsymbol{A}'_c$ and $\boldsymbol{P}'_c$ matrices. The resulting matrix $\boldsymbol{F}'_c \in \mathbb{R}^{C\times N}$ undergoes a reshape operation to obtain the final output feature matrix $\boldsymbol{F}_o \in \mathbb{R}^{C\times H\times W}$ of the TAM module. It is worth noting that the input and output features of the TAM module have the same shape size ($C \times H \times W$).

The TAM module models the correlation between sets of features that lie within distinct thresholds of the input feature map, thereby capturing the dependencies between blocks of homogeneous pixel regions. This leads to a dynamic assignment of attention to various sets of pixels, thereby enhancing the features of the input matrix.

C. Attentional Feature Enhancement Module

Our proposed AFEM comprises three branches. As depicted in Fig. 4, one branch conducts channel attention acquisition through global averaging pooling of the input feature map and two fully connected layers. Another branch, the TAM, computes the cosine similarity of the input feature map with a globally averaged feature vector. It then models correlations among similar regions to enhance the feature map with attentional features. The channel attention weights learned by the first branch are applied to the feature map obtained from TAM. The third branch, based on residual connectivity, adds the original feature maps to those from TAM and channel attention weight assignment, enabling the network to automatically learn feature assignment and facilitate gradient backpropagation. The AFEM produces a feature map of the same size as the input, rendering it easy to integrate into the network.

D. Threshold Attention Pyramid Pooling

The ASPP module employs four parallel dilation convolutions to construct features with varying perceptual fields, which enhances information extraction of objects at different scales in the image. However, this approach may lead to a loss of detailed information and insufficient global feature relevance information. To address these limitations, we propose the threshold attention pyramid pooling (TAPP) method. TAPP improves ASPP by increasing the convolution kernel size for a larger perceptual field. In addition, it adds a threshold-attention branch to model correlations between similar pixel regions, resulting in rich global contextual information with low computational cost.

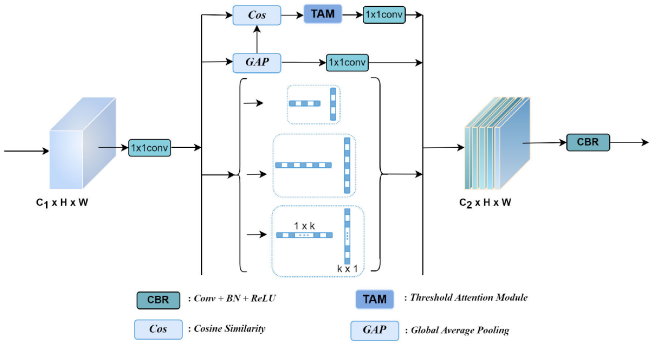


Fig. 5. Structure of the TAPP, where CBR is the convolution layer + BN layer + ReLU layer, Cos is the calculation of cosine similarity, and GAP is the calculation of GAP.

Fig. 5 shows the threshold attention space pooling module divided into three branches: expansion convolution (with varying expansion rates), global average pooling (GAP), and threshold attention. The expansion convolution branch extracts multiscale features in parallel using three dilation convolutions of sizes 4, 6, and 8 (K is both kernel size and expansion rate). To reduce computation, we use depthwise convolution with $1 \times K$ and $K \times 1$ dilation convolutions. The threshold attention branch computes input feature similarity (Cos) with GAP-computed features and inputs the results into TAM to calculate attention weights for different pixel regions. In addition, a convolution kernel size of 1 is added to both the GAP and TAM branches to reduce feature dimensionality and improve feature representation.

E. Loss Functions

For supervised training, we selected cross-entropy loss as the main predictive loss. Its formula is as follows:

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} y_k^{(n)} \log \hat{y}_k^{(n)} \tag{9}$$

where $n \in [1, 2, \ldots, N]$, N is the number of samples, and K is the number of categories. $\hat{y}_k^{(n)}$ is the one-hot vector of the network's output after softmax, and $y_k^{(n)}$ is the true label value corresponding to this sample.

Supervised training using only the difference between final layer output and true label maps slows convergence and yields limited results. To resolve this, we added an auxiliary loss in the third block of the backbone network, using cross-entropy loss as in the final layer

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda \mathcal{L}_{\mathrm{aux}} \tag{10}$$

where $\mathcal{L}_{\mathrm{CE}}$ is the prediction loss and $\mathcal{L}_{\mathrm{aux}}$ is the auxiliary loss. Hyperparameter λ balances the weights between primary and auxiliary losses. Main loss uses online hard examples mining (OHEM) to focus network learning on difficult-to-classify pixels with prediction vector probability less than θ. Hard-to-classify pixels have individual losses calculated and backpropagated for network optimization. Instead of conducting an exhaustive search for optimal parameter values,

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we determined values that produced stable segmentation effects through limited experiments on the Vaihingen dataset. Our selected parameter values were $\lambda = 0.5$, $\theta = 0.65$, and $S = 10\,000$.

IV. Datasets and Experimental Settings

A. Datasets

We evaluated the efficacy of our proposed TANet through experiments on two well-known open datasets: the International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen dataset and the ISPRS Potsdam dataset. Both datasets include six classes of remote sensing image labels: ground, building, low vegetation, tree, vehicle, and background (clutter). The ISPRS dataset provides two types of semantic labels for testing: one with eroded boundaries and one without. In our experiments, we used semantic labels with eroded boundaries.

1) Vaihingen Dataset: The ISPRS Vaihingen dataset contains 33 very high-resolution orthophoto maps. The average size of the images in the dataset is 2494 × 2064 pixels. The orthophotographs have three channels: the infrared channel, the red channel, and the green channel, each containing a wealth of spectral information. In addition, there are two sets of ancillary data in the dataset: the digital surface model (DSM) and the normalized DSM (NDSM) data. This dataset is formally divided into 16 training regions and 17 test regions. For the partitioning of the dataset, our setup is the same as [35] and [36], with the 15 images labeled as follows: 2, 4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 27, 29, 31, 33, 35, and 38 selected for training. Thirty labeled images are used for validation, and the remaining 17 images are used as the test set.

In our experiments, we did not utilize DSM or NDSM data. To train the network model, we preprocessed the remote sensing images by cropping them to 512 × 512, and data augmentation techniques were applied including random rotation (90°, 180°, and 270°), random resizing (0.5–2), the addition of random Gaussian noise, and random horizontal and vertical flipping.

2) Potsdam Dataset: The Potsdam dataset comprises 38 fine-resolution images, all 6000 × 6000 pixels in size. The dataset includes near infrared (NIR), red, green, and blue channels, as well as DSM and NDSM data. We divided it in the same way as [36], using the 14 images labeled: 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, 4_15, 5_13, 5_14, 5_15, 6_13, 6_14, 6_15, and 7_13 for testing, ID: 2_10 for validation, and except for image 7_10 with incorrect annotations, 22 images were utilized as the training set. As with the Vaihingen dataset, we did not use the DSM and NDSM data. We used image cutting and data enhancement in the same way as on the Vaihingen dataset.

B. Evaluation Metrics

The performance of TANet was evaluated using three metrics: overall accuracy (OA), mean intersection over union (mIoU), and F1-score. Based on the cumulative confusion matrix, these evaluation metrics are calculated as

$$\mathrm{OA} = \frac{\sum_{k=1}^{N} \mathrm{TP}_k}{\sum_{k=1}^{K} \mathrm{TP}_k + \mathrm{FP}_k + \mathrm{TN}_k + \mathrm{FN}_k} \tag{11}$$

TABLE I
Results of Ablation Experiments on the Vaihingen Dataset for the L_1 Parameter in the AFEM Module

L_1	50	100	150	200	250	300
Mean F1(%)	90.54	90.66	**90.78**	90.64	90.50	90.51
OA(%)	90.97	90.94	**91.13**	91.03	90.77	90.90
mIoU(%)	82.94	83.17	**83.35**	83.15	82.91	82.90

$$\mathrm{mIoU} = \frac{1}{N} \sum_{k=1}^{N} \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FP}_k + \mathrm{FN}_k} \tag{12}$$

$$\mathrm{precision}_k = \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FP}_k} \tag{13}$$

$$\mathrm{recall}_k = \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FN}_k} \tag{14}$$

$$\mathrm{F1}_k = 2 \times \frac{\mathrm{precision}_k \times \mathrm{recall}_k}{\mathrm{precision}_k + \mathrm{recall}_k} \tag{15}$$

where TP_k, TN_k, FN_k, and FP_k denote true positives, false positives, true negatives, and false negatives, respectively, for a particular object indexed as category k.

C. Implementation Details

For all comparisons, we employ ResNet-101 pretrained on the ImageNet dataset as the backbone network. The final two downsampling operations are replaced with dilated convolutional layers with expansion rates of 2 and 4 [37], resulting in an output stride of 8. The AdamW optimizer, which includes weight decay, is used. During training, a "poly" strategy is applied to set the learning rate, calculated as the initial learning rate multiplied by $(1 - (\mathrm{max_iter}/\mathrm{iter}))^{0.9}$, with an initial value of 0.0005. Experiments were conducted on an NVIDIA Tesla V100 GPU with 32-GB memory. The threshold number L of the threshold attention module was optimized for different datasets.

V. Experimental Results and Analysis

A. Parameter Study for the TANet

The proposed threshold attention module has a crucial parameter, L, referred to as the threshold number. This parameter determines the level of granularity in the attention applied to the input features. We experimentally studied the effect of L on the segmentation performance of the network. We studied the effect of threshold values L_1 and L_2 in the AFEM and TAPP modules on the Vaihingen and Potsdam datasets, respectively. We first set L_2 in TAPP to 200 and sought the optimal value of L_1 in AFEM. Next, we found the optimal value of L_2. In addition, we evaluated the impact of incorporating dilated convolutions of varying scales into the TAPP module on the model's segmentation performance via an experiment.

1) Experiments on the Vaihingen Dataset: The results of the experiment on the Vaihingen dataset are presented in Tables I and II. It is evident that TANet obtains the optimal semantic segmentation performance when L_1 is set to 150 and L_2 is set to 200. Furthermore, it is observed that the model demonstrates a greater sensitivity to the parameter

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TABLE II
RESULTS OF ABLATION EXPERIMENTS ON THE VAIHINGEN DATASET FOR THE L_2 PARAMETER IN THE TAPP MODULE

L_2	50	100	150	200	250	300
Mean F1(%)	90.37	90.51	90.69	**90.78**	90.64	90.45
OA(%)	90.77	91.13	90.97	**91.13**	91.11	91.07
mIoU(%)	82.67	82.91	83.22	**83.35**	83.13	82.81

TABLE III
RESULTS OF ABLATION EXPERIMENTS ON THE VAIHINGEN DATASET FOR DILATED CONVOLUTION IN TAPP

Method	4x4	6x6	8x8	Mean F1(%)	OA(%)	mIoU(%)
TANet				90.57	90.90	82.98
TANet	✓			90.62	90.93	83.11
TANet	✓	✓		90.71	91.10	83.24
TANet	✓	✓	✓	**90.78**	**91.13**	**83.35**

TABLE IV
RESULTS OF ABLATION EXPERIMENTS ON THE POTSDAM DATASET FOR THE L_1 PARAMETER IN THE AFEM MODULE

L_1	50	100	150	200	250	300
Mean F1(%)	93.16	93.19	93.30	**93.35**	93.29	93.13
OA(%)	91.79	91.85	91.94	**92.10**	91.93	91.63
mIoU(%)	87.43	87.48	87.68	**87.75**	87.64	87.41

L_2 compared with L_1. Table III displays the outcomes of the ablation experiments on the Vaihingen dataset for the dilated convolution in TAPP. The results reveal that the use of all three scales of the dilated convolution enhances segmentation performance, with optimal results obtained when all three scales are utilized together.

2) Experiments on the Potsdam Dataset: The results of the Potsdam dataset are presented in Tables IV and V. The optimal semantic segmentation is achieved when both L_1 and L_2 are set to 200. No significant difference in the sensitivity to parameters L_1 and L_2 was observed. Similar to the results obtained on the Vaihingen dataset, all three scales of the dilated convolution in Table VI help to improve the final segmentation performance of TANet.

B. Ablation Study

1) Comparison With Context Aggregation Modules and Attention Modules: Table VII compares our proposed model with classical context extraction modules and four newer attention mechanisms in terms of segmentation effectiveness. The results indicate that the AFEM and TAPP modules achieve better segmentation accuracy than other context extraction modules and attention mechanisms. Our proposed thresholded attention results in a more effective extraction of attentional information by modeling the feature correlation among different pixel regions. The experimental results proved the effectiveness of the method.

The improvement effect of the TAPP module and the AFEM module on the network model is comparable. Both TAPP and AFEM modules effectively enhance the semantic segmentation performance of the model. The combination of these two

TABLE V
RESULTS OF ABLATION EXPERIMENTS ON THE POTSDAM DATASET FOR THE L_2 PARAMETER IN THE TAPP MODULE

L_2	50	100	150	200	250	300
Mean F1(%)	93.13	93.17	93.16	**93.35**	93.15	93.13
OA(%)	91.79	91.87	91.79	**92.10**	91.76	91.80
mIoU(%)	87.39	87.45	87.45	**87.75**	87.41	87.38

TABLE VI
RESULTS OF ABLATION EXPERIMENTS ON THE POTSDAM DATASET FOR DILATED CONVOLUTION IN TAPP

Method	4x4	6x6	8x8	Mean F1(%)	OA(%)	mIoU(%)
TANet				93.04	91.54	87.22
TANet	✓			93.10	91.70	87.34
TANet	✓	✓		93.28	91.99	87.65
TANet	✓	✓	✓	**93.35**	**92.10**	**87.75**

TABLE VII
RESULTS OF THE ABLATION EXPERIMENTS ON THE VAIHINGEN DATASET

Method	Mean F1(%)	OA(%)	mIoU(%)
ResNet-101 Baseline	89.84	90.38	81.82
ResNet-101+SE [31]	90.11	90.80	82.26
ResNet-101+SA [20]	90.16	90.70	82.30
ResNet-101+ASPP [36]	90.23	90.88	82.46
ResNet-101+DAB [35]	90.33	90.97	82.60
ResNet-101+PPM [37]	90.34	90.80	82.62
ResNet-101+EA [30]	90.41	90.77	82.75
ResNet-101+CAM&KAM [34]	90.41	90.96	82.71
ResNet-101+BCM&CEM [24]	90.44	90.82	82.76
ResNet-101+CAA&RSA [19]	-	90.98	82.87
ResNet-101+TAM	90.58	90.90	83.03
ResNet-101+AFEM	90.63	91.09	83.14
ResNet-101+TAPP	90.65	90.99	83.15
ResNet-101+TAPP&AFEM (ours)	**90.78**	**91.13**	**83.35**

modules and the baseline network results in TANet, which achieves the best segmentation results.

2) Efficiency Comparison: We compare the efficiency of our proposed AFEM and TAPP modules with other contextual aggregation and attention modules in terms of parameters, GPU memory, and computational costs (GFLOPs). To ensure a fair comparison, as in [34] and [19], we use 3 × 3 convolutions for dimensionality reduction and evaluate the cost without considering the backbone cost. Table VIII displays the experimental results. Compared with the standard SA mechanism, the proposed module exhibits approximately 1/12 the GFLOPs, 1/2 the number of model parameters, and 1/8 the GPU memory. The optimized GFLOPs and the number of model parameters demonstrate the superiority of our AFEM and TAPP modules compared with the state-of-the-art methods.

Table IX presents the time cost of the model in the inference phase. In this experiment, the backbone of all models is set to ResNet-101. While our method may not have the most optimal time-cost performance, the time required by our model is not significantly different from other models. For instance, TANet required only 5 ms more than EANet and 3 ms more than DeepLabV3+. We consider the slight increase in time

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TABLE VIII

EFFICIENCY COMPARISON WITH CONTEXT AGGREGATION MODULES AND ATTENTION MODULES WHEN PROCESSING INPUT FEATURE MAP OF SIZE ($1 \times 2048 \times 128 \times 128$) DURING THE INFERENCE STAGE

Method	GFLOPs	Params(M)	Memory(MB)
LKPP [1]	884	54.5	818
PPM [37]	619	22.0	792
ASPP [36]	503	15.1	284
CCA [38]	804	10.6	427
SA [20]	619	10.5	2168
OCR [39]	354	10.5	202
CAA&RSA [19]	292	13.1	393
PAM&AEM [15]	158	10.4	489
CAM&KAM [34]	86	5.3	**160**
AFEM&TAPP (ours)	**49**	**4.5**	262

TABLE IX

RESULTS OF INFERENCE TIME COMPARISON BETWEEN TANET AND OTHER MODELS

Method	Average inference time per image (seconds)
FCN [9]	0.071 ± 0.001
SA [20]	0.076 ± 0.001
PSPNet [37]	0.084 ± 0.001
EANet [30]	0.086 ± 0.001
SENet [31]	0.088 ± 0.001
DeeplabV3+ [36]	0.088 ± 0.001
DABNet [35]	0.097 ± 0.001
TANet (ours)	0.091 ± 0.001

spent to be a reasonable tradeoff for obtaining improved segmentation results and significantly reducing the number of model parameters.

C. Comparison With the State of the Art

1) Experimental Results on the Vaihingen Dataset: Similar to HMANet [19] and HBCNet [24], our proposed methods aim to enhance the model's segmentation results. Two improvements were made to the Loss function to enhance TANet network segmentation. The first is the Aux Loss, which accelerates convergence and improves segmentation outcomes. The second is the OHEM method, which focuses the network more on challenging-to-classify pixels. We also employed the technique of test-time enhancement (TTA). TTA involves flipping the input images horizontally and vertically during testing, leading to improved segmentation performance of the model. Our findings are presented in Table X and demonstrate that all three methods effectively improve the model's segmentation ability.

Table XII shows a comparison of our best segmentation results on the Vaihingen test set with the state-of-the-art methods, including contextual aggregation methods and various attention-based methods. Our TANet uses ResNet-101 as the backbone, like most models. The results reveal that TANet outperforms the other methods, achieving the best results in all three important composite metrics. The experimental result supports the efficacy of our TAM and TANet architecture. We also experimented with adding the AFEM and TAPP modules to the ResNet50 and VGG-19 backbones. The experimental results in Tables XII and XIII show that the addition of the AFEM and TAPP modules on other different backbones can also effectively improve the segmentation of the model. The "-" symbol in the tables throughout this article signifies the absence of data provided by the authors of the original article. In addition, reproducing their network model is challenging as the underlying code is not available as open source.

TABLE X

RESULTS OF ABLATION EXPERIMENTS WITH DIFFERENT IMPROVEMENTS ON THE VAHINGEN DATASET

Method	OHEM	Aux Loss	TTA	Mean F1(%)	OA(%)	mIoU(%)
TANet				90.78	91.13	83.35
TANet	✓			90.85	91.23	83.46
TANet	✓	✓		91.16	91.50	83.99
TANet	✓	✓	✓	**91.45**	**91.93**	**84.45**

TABLE XI

RESULTS OF ABLATION EXPERIMENTS WITH DIFFERENT IMPROVEMENTS ON THE POTSDAM DATASET

Method	OHEM	Aux Loss	TTA	Mean F1(%)	OA(%)	mIoU(%)
TANet				93.35	92.10	87.75
TANet	✓			93.40	92.27	87.85
TANet	✓	✓		93.52	92.32	88.06
TANet	✓	✓	✓	**93.71**	**92.95**	**88.43**

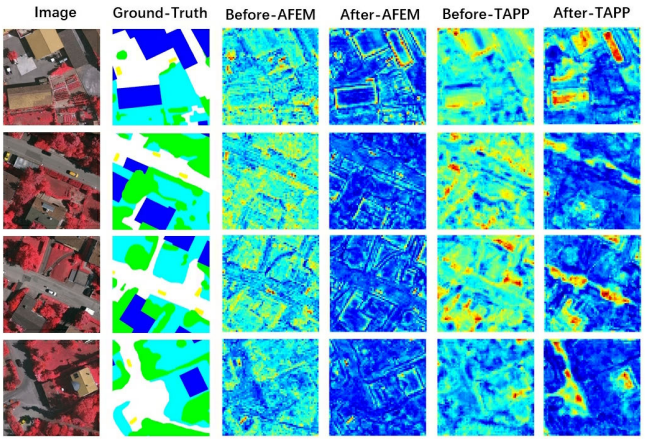


Fig. 6. Visualization of the AFEM module and TAPP module input and output features plotted on the Vaihingen test set.

2) Visualization of the Attention Module: To enhance the comprehension of the roles of the AFEM and TAPP modules, which were designed based on TAM, the feature maps before and after these modules were visualized. The results are presented in Fig. 6. The AFEM module enhances edge differences between pixel blocks belonging to different objects, making object contours clearer and preserving more detailed information.

Comparing the before-TAPP and after-TAPP columns, it can be seen that the TAPP module enhances response value differences between regions belonging to different objects, making it easier for the model to distinguish semantic information. For instance, response values are relatively larger for both buildings and cars.

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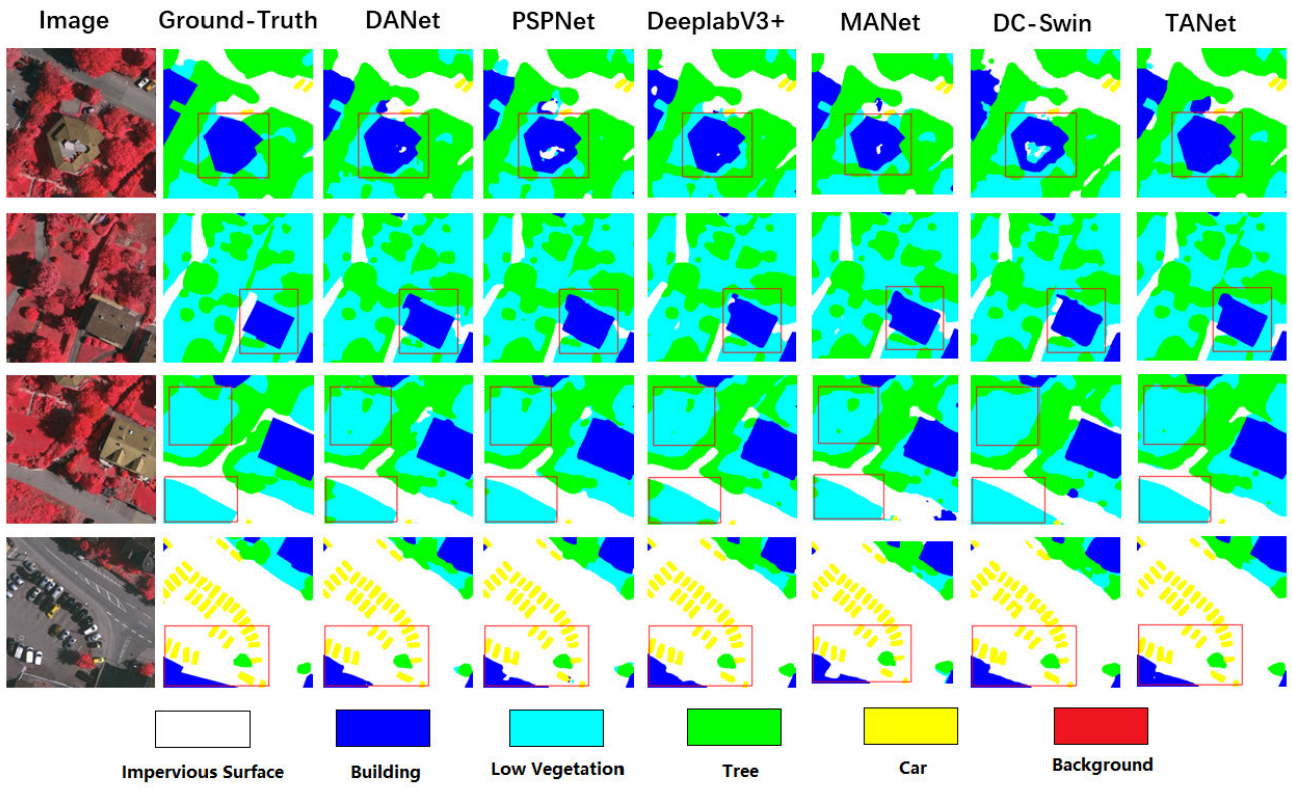


Fig. 7. Qualitative comparison between our method (TANet) and other methods on the Vaihingen dataset. The region in the red box represents a challenging segmentation area.

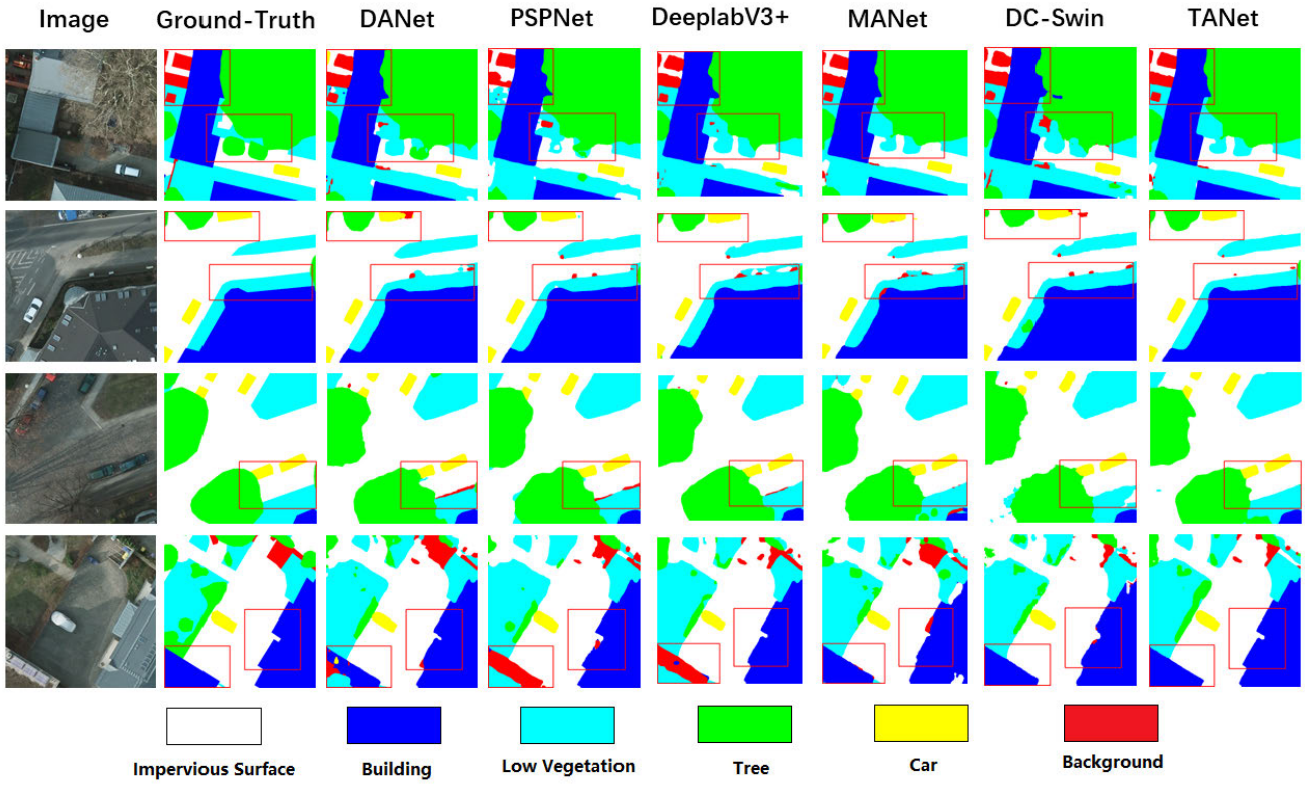


Fig. 8. Qualitative comparison between our method (TANet) and other methods on the Potsdam dataset. The region in the red box represents a challenging segmentation area.

3) Visualization of Results: As shown in Fig. 7, we visualize the segmentation results of TANet on the Vaihingen test set and compare them qualitatively with several classical semantic segmentation networks. The region in the red box represents a challenging segmentation area. A comparison of the models' results clearly shows that TANet's predictions are the most

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TABLE XII
QUANTITATIVE COMPARISONS WITH STATE OF THE ARTS ON THE VAIHINGEN TEST SET

	Backbone	Imp.surf	Building	Low veg	Tree	Car	Mean F1(%)	OA(%)	mIoU(%)
V-FuseNet [40]	-	92.00	94.40	84.50	89.90	86.30	89.42	90.00	-
DLR_9 [41]	-	92.40	95.20	83.90	89.90	81.20	88.52	90.30	-
TreeUNet [42]	-	92.50	94.90	83.60	89.60	85.90	89.30	90.40	-
DANet [35]	ResNet-101	91.63	95.02	83.25	88.87	87.16	89.19	90.44	81.32
DeepLabV3+ [36]	ResNet-101	92.38	95.17	84.29	89.52	86.47	89.57	90.56	81.47
ABCNet [43]	ResNet-18	92.70	95.20	84.50	89.70	85.30	89.50	90.70	81.30
PSPNet [37]	ResNet-101	92.79	95.46	84.51	89.94	88.61	90.26	90.85	82.58
ACFNet [44]	ResNet-101	92.93	95.27	84.46	90.05	88.64	90.27	90.90	82.68
MANet [34]	ResNet-101	93.02	95.47	84.64	89.98	88.95	90.41	90.96	82.71
CASIA2 [45]	ResNet-101	93.29	96.00	84.70	89.90	86.70	90.10	91.10	-
CCANet [38]	ResNet-101	93.29	95.53	85.06	90.34	88.70	90.58	91.11	82.76
HMANet [19]	ResNet-101	93.50	95.86	85.41	90.40	89.63	90.96	91.44	83.49
MFNet [46]	ResNet-50	93.43	96.35	85.85	90.50	88.31	90.88	91.67	83.50
CTMFNet [25]	HRNet&transformer	93.79	96.12	85.02	90.47	**91.47**	91.37	91.60	84.34
DC-Swin [47]	Swin-S	93.60	96.18	85.75	90.36	87.64	90.71	91.63	83.22
HBCNet [24]	HRNet_w48	93.60	96.13	85.95	**90.53**	90.40	91.32	91.72	84.21
TANet (Ours)	VGG-19	93.21	96.23	85.01	88.09	90.04	90.52	90.88	82.91
TANet (Ours)	ResNet-50	93.79	96.64	85.77	88.21	90.84	91.05	91.37	83.81
TANet (Ours)	ResNet-101	**94.16**	**96.80**	**86.95**	88.84	90.52	**91.45**	**91.93**	**84.45**

TABLE XIII
QUANTITATIVE COMPARISONS WITH STATE OF THE ARTS ON THE POTSDAM TEST SET

	Backbone	Imp.surf	Building	Low veg	Tree	Car	Mean F1(%)	OA(%)	mIoU(%)
UZ_1 [48]	-	89.30	95.40	81.80	80.50	86.50	86.70	85.80	-
DANet [35]	ResNet-101	91.96	96.35	86.20	87.21	95.92	91.48	89.98	84.57
V-FuseNet [40]	-	92.70	96.30	87.30	88.50	95.40	92.04	90.60	-
TSMTA [49]	ResNet - 101	92.91	97.13	87.03	87.26	95.16	91.90	90.64	-
TreeUNet [42]	-	93.10	97.30	86.60	87.10	95.80	91.98	90.70	-
DeepLabV3+ [36]	ResNet - 101	92.95	95.88	87.62	88.15	96.02	92.12	90.88	84.32
CASIA3 [45]	ResNet - 101	93.40	86.80	87.60	88.30	96.10	92.44	91.00	-
PSPNet [37]	ResNet - 101	93.36	96.97	87.75	88.50	95.42	92.40	91.08	84.88
MANet [34]	ResNet - 50	93.40	96.96	88.32	89.36	96.48	92.90	91.32	86.95
CCANet [38]	ResNet - 101	93.58	96.77	86.87	88.59	96.24	92.41	91.47	85.65
HUSTW4 [50]	-	93.60	97.60	88.50	88.80	94.60	92.62	91.60	-
MFNet [46]	ResNet - 50	94.25	97.52	88.42	89.43	96.62	93.25	91.96	87.57
HMANet [19]	ResNet - 101	93.85	97.56	88.65	89.12	96.84	93.20	92.21	87.28
CTMFNet [25]	HRNet&transformer	93.22	97.12	87.87	89.36	96.61	92.84	91.38	86.85
HBCNet [24]	HRNet_w48	94.29	97.54	88.49	89.58	97.00	93.38	91.97	87.81
DC-Swin [47]	Swin-S	94.19	97.57	88.57	**89.62**	96.31	93.25	92.00	87.56
TANet (Ours)	VGG-19	93.99	97.42	88.95	87.68	97.05	93.02	91.20	87.22
TANet (Ours)	ResNet-50	94.50	97.57	89.61	88.72	97.50	93.58	91.93	88.17
TANet (Ours)	ResNet-101	**94.65**	**97.65**	**89.80**	88.97	**97.54**	**93.72**	**92.45**	**88.41**

similar to the true labeled maps in terms of object consistency and boundary definition. This emphasizes the effectiveness of the TAM in modeling pixel region features and enhancing object boundary details.

4) Experimental Results on the Potsdam Dataset: To further assess the efficacy of TANet, experiments were conducted on the Potsdam dataset using the same three methods listed in Table XI as those performed on the Vaihingen dataset. Results were compared with the latest available methods and are shown in Table XIII. TANet achieved the highest scores in the three metrics of average F1, OA, and mIoU, outperforming all other models. Our method outperformed other approaches in most categories, with the exception of the Tree category. Further analysis suggests that this may be due to the thin branches and wide color distribution of the Tree category in the two datasets. These characteristics may make it difficult for our threshold attention method to accurately detect the region associated with this category. Fig. 8 visualizes TANet's segmentation results on the Potsdam test set, with the closest prediction to ground truth indicated in the red-boxed area.

VI. CONCLUSION

In this article, we propose a novel TAM. In comparison with SAs, TAM significantly reduces computational effort while augmenting the correlation modeling between similar pixel regions in the feature map. Based on TAM, we design TANet, a semantic segmentation network for remote sensing images. TANet employs a pretrained ResNet-101 as the backbone and extracts global relevance feature information from the deep network using the TAPP module. The shallow network output

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is augmented with region-specific feature information via the AFEM module, and the complementary information from both is subsequently combined to obtain the final prediction map. To validate our approach, we conducted experiments on two high-resolution remote sensing image semantic segmentation datasets: Vaihingen and Potsdam. The results show that TANet outperforms other methods in most overall metrics, demonstrating the efficacy of our approach.

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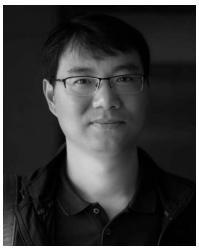
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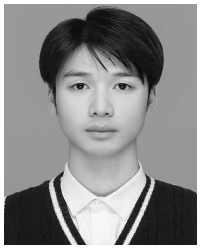
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